

Data C100 - Final

Summer 2026

Write your name BIG and clearly: _____

Email: _____@berkeley.edu

Student ID: _____

Name and SID of the person on your left: _____

Name and SID of the person on your right: _____

Exam Room: _____ Seat Number: _____

Instructions:

This exam consists of **71 points** spread out over **9 questions**. The exam must be completed in **minutes** unless you have accommodations supported by a DSP letter.

- Note that some questions have circular bubbles to select a choice. Please shade in the circle fully to mark your answer.
- **Write clearly and legibly.** We reserve the right to withhold points from answers that are very difficult to read.
- Blank answers and incorrect answers are graded identically, so it's in your best interest to answer every question.
- **You MUST write your Student ID number at the top of each page.**
- You should not use a calculator, scratch paper, or notes you own other than the reference sheets distributed at the beginning of the exam.

For all Python questions, you may assume Pandas has been imported as `pd`, NumPy as `np`, the Python RegEx library as `re`, `matplotlib.pyplot` as `plt`, and `seaborn` as `sns`.

Honor Code:

As a member of the UC Berkeley community, I act with honesty, integrity, and respect for others. I am the person whose name is on the exam, and I completed this exam in accordance with the Honor Code.

Signature: _____

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1 Potpourri [8 Pts]

- (a) [2 Pts] Table A has 200 rows. Table B has 300 rows. The ID column of A has 200 unique values. The ID column of table B has 300 unique values. The tables are joined on the ID column.

Complete the table below by writing the minimum and maximum number of rows that could result from each join.

Join Type	Minimum Number of Rows	Maximum Number of Rows
Inner join of A and B		
Left join of A to B		
Left join of B to A		
Full join of A and B		

Solution:	Join Type	Minimum Rows	Maximum Rows
	Inner join of A and B	0	200
	Left join of A to B	200	200
	Left join of B to A	300	300
	Full join of A and B	300	500

- (b) [2 Pts] Let `df` be the following DataFrame:

a	b
10	10
40	30
30	20
20	5
5	5

`z` is a boolean Series with the values `[True, False, False, True, False]`

Write the output of the following. If an answer results in a series, write the values of the series in the order they appear.

Code	Output
<code>np.sum(df[z]['a'])</code>	
<code>np.mean(z)</code>	
<code>np.sum(df[2:3]['b'])</code>	
<code>np.sum(df['a'] == df['b'])</code>	

Solution:	Code	Output	
	<code>np.sum(df[z]['a'])</code>	30	
	<code>np.mean(z)</code>	2/5	
	<code>np.sum(df[2:3]['b'])</code>	20	
	<code>np.sum(df['a'] == df['b'])</code>	2	

(c) [4 Pts] For each combination of variables below, write three plot types that could be used to visualize the variables in a single plot.

- You must use plot types that we've taught in Data 100.
- If you write more than three plot types for a variable, you will lose credit.
- If there are only two possible answers for a question, write NA as the third plot type.

Variable type(s)	Plot Type 1	Plot Type 2	Plot Type 3
One quantitative			
Two quantitative			
One quantitative and one qualitative nominal			
Two quantitative and one qualitative ordinal			

Solution:

- **1 quantitative variable**
 - Histogram
 - KDE
 - Boxplot
 - Rug plot
 - Scatterplot (technically correct)
- **2 quantitative variables**
 - Line plot
 - Hexplot
 - Contour plot
 - Scatterplot
- **1 quantitative variable, 1 qualitative nominal variable**
 - Side-by-side boxplots
 - Overlaid histograms
 - Overlaid KDEs

- Barplot (If one variable is a category and the other is the count of that category)
- Scatterplot (technically correct)
- **2 quantitative variables, 1 qualitative ordinal variable**
 - Line plot
 - Scatterplot
 - NA is an acceptable third answer, though someone could also answer with overlaid contour plot.

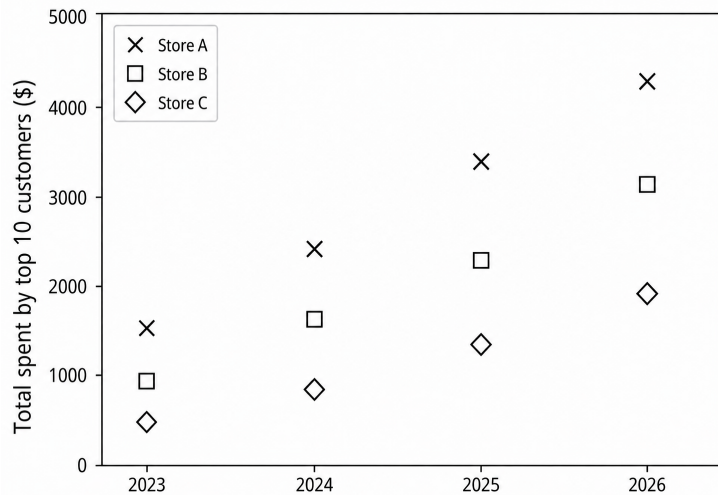
2 EDA [10 Pts]

A company has three stores: Store A, Store B, and Store C. Every year, the company identifies the 10 customers with the highest total **count** of transactions combined across all its stores in that year.

- For example, in 2024, suppose a customer makes 3 transactions at Store A, 2 transactions at Store B, and 4 transactions at store C. The company would compare this customer's 9 total transactions against the total number of transactions of all other customers in 2024.

Then, the company computes the total **dollar amount** that the top 10 customers spent at each of the company's stores in the year the customers were selected.

Consider the following visualization of this data:



The visualization above was generated with a single call to `sns.scatterplot` and a single DataFrame called `plot_df`.

(a) [1 Pt] How many rows are in `plot_df`?

12

(b) [1 Pt] What is the minimum number of columns in `plot_df`?

3

(c) [8 Pts] `plot_df` was generated by transforming a different DataFrame called `raw_df`. Each row of `raw_df` represents one transaction. Here are the columns of `raw_df`:

- `transaction_id`: a unique identifier for the transaction
- `customer_id`: an identifier for the customer associated with the transaction
- `store_id`: an identifier for the store where the transaction occurred
- `year`: the year of the transaction
- `amount`: the dollar amount of the transaction

Write an analysis plan to transform `raw_df` into `plot_df`. Things to keep in mind:

- The same customer might be in the top 10 for multiple years.
- You can assume the company had at least 10 unique customers each year.
- `plot_df` must only include rows that are needed for the plot. No extra rows are allowed.
- `plot_df` may optionally include columns that are not required to make the plot, since the call to `sns.scatterplot` could ignore these columns.
- Make sure to use the appropriate canonical English terms we learned in lecture. See the reference sheet for examples of these terms.

Write your answer on the next page! →

Repeated descriptions: Every year, the company identifies the 10 customers with the highest total **count** of transactions combined across all of its stores in that year. Then, the company computes the total **dollar amount** that the top 10 customers spent at each of the company's stores in the year the customers were selected.

Reminders: `raw_df` contains transaction ID, customer ID, store ID, year, and amount. The plot shows year on the x-axis and total amount spent by the top 10 customers on the y-axis. The point shape corresponds to the store. You are writing steps to produce `plot_df`.

Hint: You may find it helpful to produce multiple *DataFrames* and then join them. Our solution produces two *DataFrames* and joins them.

To be eligible for partial credit, keep your answer organized and neat.

4. Sum the total amounts across the 10 customers for each store ID and year.

Solution:

- 1a. Starting from `raw_df`, group by year and customer ID. Count the rows in each group.
 - This produces a DataFrame with the number of transactions made by each customer in each year.
- 1b. Sort the resulting DataFrame in descending order of number of transactions. Group the resulting DataFrame by year. Select the first 10 rows in each group.
 - This produces a DataFrame containing the 10 customer IDs with the most transactions in each year. There is also a column with the number of transactions, but we can ignore it.
2. Starting from `raw_df` again, group by store ID, customer ID, and year. Sum the amount column for each group.
 - Now you have a DataFrame with total amount spent by each customer ID during each year at each store.
3. Left or inner join the first DataFrame to the second, joining by customer ID and year.
 - While this step could technically be implemented as a filter, it is much simpler to implement it as a join. To receive credit for a filter, it must be described precisely. For example, a customer may be in the top 10 in one year, but not another year. The filter must take into account both customer ID and year, not just customer ID.

3 I still love all of my regexes [4 Pts]

Hazel obtains text from the "Staff" page of the Data 100 website. She wants to use RegEx to extract certain pieces of information.

Here is an example string. The character "_" represents a single space, and you may use it in your response.

```
"Name:_Oski;_Age:_161;_Courses_taken:_D100,
_DATA143,_CS61B,_Math54,_Data27s;_Phone Number:
_510-642-9999;_Likes:_[Bears,_Waffles,_Data Science]"
```

```
pattern = r"[A-Z][a-z]+\W*\d{1,3}\w?"
```

List matches of the pattern above in the order they appear in the string, with the first match next to Match 1.

Note: You may have more or fewer than 6 total matches. For any unmatched slots, **write** No Match **instead**. If there are more than six matches, just write the first six that appear.

Match 1: '_____'

Match 2: '_____'

Match 3: '_____'

Match 4: '_____'

Match 5: '_____'

Match 6: '_____'

Solution:

Match 1: 'Age: 161'
Match 2: 'Math54'
Match 3: 'Data27s'
Match 4: 'Number: 510'
Match 5: 'No Match'
Match 6: 'No Match'

If last 2 Matches left blank, it is still a valid solution.

4 Sampling [4 Pts]

Sarah is the data scientist who oversees the **BayPass Program**. The BayPass Program is a study that analyzes the effects of giving UC Berkeley students free access to all Bay Area transportation services through a BayPass transit card.

- (a) Sarah takes a simple random sample of 100 UC Berkeley students from the UC Berkeley enrolled student database and gives them each a BayPass transit card. She collects usage data from these cards for analysis.

(i) [1 Pt] What is the population of interest of the study?

- ☐ All BayPass card holders
☒ **All UC Berkeley students**
☐ All Bay Area students who use Bay Area transit services
☐ All enrolled UC Berkeley students who use Bay Area transit services

Solution: Sarah wants to study "effects of providing *UC Berkeley students*" with...", so the population of interest is all UC Berkeley students.

(ii) [1 Pt] What is the sampling frame in Sarah's study? Be specific.

Solution: All students in the UC Berkeley enrolled student database

- (b) [2 Pts] Sarah considers using a stratified random sample to select 100 students, instead of a simple random sample. Why might Sarah want to consider a stratified random sample?

- ☒ **True** ☐ False She wants to guarantee that small but important groups are represented in the sample.
☐ True ☒ **False** She wants to reduce selection bias.
☐ True ☒ **False** She wants to reduce reporting bias (i.e., response bias).
☒ **True** ☐ False She wants to reduce chance error.
☐ True ☒ **False** She wants to reduce non-response bias.

Solution: Stratified random sampling ensures representation from particular groups, which makes the sample less variable across possible samples that could be obtained. Chance error goes down when a sample is less variable.

5 Shell We Do Some Regression? [6 Pts]

Zara is building a model to predict how long it takes a turtle to finish a race, `finish_time` (in minutes), using features like `aura` (a turtle's charisma score) and `shell_diameter` (in cm).

(a) [1.5 Pts] We now use the following model equation to predict `finish_time`:

$$\text{finish_time} = \hat{\theta}_0 + \hat{\theta}_1 \times \text{aura}$$

For each action, determine whether it would change the $\hat{\theta}_1$ value.

- ☐ True ☒ **False** Adding a constant to all values of `aura`.
☒ **True** ☐ False Multiplying all values of `aura` by 3.
☒ **True** ☐ False Using the square root of `aura` as the predictor variable.

Solution: Recall that the SLR formula for $\hat{\theta}_1$ in a model equation $\hat{y} = \hat{\theta}_0 + \hat{\theta}_1 x$ is

$$\hat{\theta}_1 = r \frac{\sigma_y}{\sigma_x}.$$

The first statement is false. Adding a constant to a predictor variable changes neither its standard deviation nor its correlation with the response variable.

The second statement is true. Multiplying the predictor by 3 scales σ_x by 3 without changing r , so $\hat{\theta}_1$ changes.

The third statement is true. Replacing the predictor with its square root generally changes both its standard deviation and its correlation with the response.

(b) [3 Pts] We now use the following model equation to predict `finish_time`:

$$\text{finish_time} = \hat{\theta}_0 + \sum_{j=1}^d \hat{\theta}_j \times \text{aura}^j$$

and train it with L2 regularization and hyperparameter λ . We perform 5-fold cross-validation and find the following table of cross-validation errors for each (λ, d) pair:

	$\lambda = 0.02$	$\lambda = 0.08$	$\lambda = 0.15$	$\lambda = 0.30$
$d = 1$	6.8	3.9	1.5	2.1
$d = 2$	5.9	0.7	2.4	4.0
$d = 3$	1.3	4.1	5.2	6.9

- (i) [1 Pts] How many times do we need to evaluate a model on a validation set to construct the above table?

Solution: There are 4 values of λ and 3 values of d , and 5-fold cross-validation fits and evaluates a model 5 times for each (λ, d) pair, so the answer is $4 \times 3 \times 5 = 60$.

- (ii) [1 Pts] Which pair of hyperparameters (λ, d) in the table above is expected to lead to the highest model bias squared?

Solution: $(\lambda = 0.30, d = 1)$

- (iii) [1 Pts] Which pair of hyperparameters (λ, d) in the table above is expected to lead to the highest model variance?

Solution: $(\lambda = 0.02, d = 3)$

Solution: For part (i), provided there are 4 options for λ , 3 options for d , and 5-fold cross-validation, the answer is $4 \times 3 \times 5 = 60$.

For parts (ii) and (iii), note that a higher λ imposes more regularization and therefore higher model bias, while a higher d provides more model features and therefore higher model variance. Therefore, part (ii) has answer $(\lambda = 0.30, d = 1)$ (highest λ , fewest features), and part (iii) has answer $(\lambda = 0.02, d = 3)$ (lowest λ , most features) — regardless of the specific numbers shown in the table.

- (c) [1.5 Pts] We now train the following model to predict `finish_time`:

$$\text{finish_time} = \hat{\theta}_0 + \hat{\theta}_1 \times (\text{shell_diameter})^2$$

We find the following optimal parameters for the model: $\hat{\theta}_0 = 2$, $\hat{\theta}_1 = 2$. We evaluate our model on the following dataset:

	shell_diameter	finish_time
Michaelangelo	1	1
Raphael	2	9
Donatello	3	13

What is the empirical risk if we fit the model with **L1 loss** and **L1 regularization** with hyperparameter $\lambda = \frac{2}{3}$? The intercept is not regularized. You can leave your answer as an unsimplified algebraic expression with numbers plugged in.

Solution: Note that we don't regularize the intercept of the model. Since we're using L1 loss and L1 regularization, the empirical risk is:

$$\begin{aligned}
 R(\vec{\theta}) &= \left(\frac{1}{3} \sum_{i=1}^3 |\hat{y}_i - y_i| \right) + \lambda |\hat{\theta}_1| \\
 &= \frac{1}{3} \left[|(2 + 2 \times 1^2) - 1| + |(2 + 2 \times 2^2) - 9| + |(2 + 2 \times 3^2) - 13| \right] + \lambda |\hat{\theta}_1| \\
 &= \frac{1}{3} [3 + 1 + 7] + \frac{2}{3} \times 2 \\
 &= \frac{1}{3}(11) + \frac{4}{3} = \frac{11}{3} + \frac{4}{3} = \frac{15}{3} = 5.
 \end{aligned}$$

6 Spiderman: Far From the Minimum [13 Pts]

Kevin wants to estimate how long it takes Spiderman to get to an emergency, `response_time` (in minutes), using a dataset of Spiderman's past emergency responses, `spidey_calls`.

- (a) [2 Pts] Kevin is interested in creating a constant model ($\hat{y} = \theta_0$) to make predictions. He first must decide between optimizing MSE or MAE. Select all statements below that are true.
- ☐ It is possible for two different constant predictions to both result in the optimal MAE.
 - ☐ Since the MSE loss surface is always convex and smooth, there is always one optimal θ_0 that minimizes it.
 - ☐ Kevin can use gradient descent to find a value of θ_0 that minimizes MAE.
 - ☐ Suppose Kevin chooses to optimize the MSE. The median of the data could result in a lower training loss than the training loss associated with the mean.

Solution: The first statement is true: for a constant model, the MSE loss surface is a smooth, convex parabola in θ_0 with a single minimizer.

The second statement is true: with an even number of data points, any value between the two middle order statistics achieves the same optimal MAE.

The third statement is true: even though the MAE loss surface isn't always smooth, gradient descent can still land on the optimal constant model.

The fourth statement is false: the mean is exactly optimal for MSE, so no other choice (including the median) can beat it.

- (b) [1 Pt] Suppose Kevin's design matrix $\mathbb{X}$ has full column rank, so he generates unique OLS predictions $\hat{\mathbb{Y}}$. Which of the following is true?

- ☐ True ☒ **False** The residual vector $\mathbb{Y} - \hat{\mathbb{Y}}$ has the same dimension as the parameter vector $\vec{\theta}$.
- ☒ **True** ☐ False The predictions $\hat{\mathbb{Y}}$ are the projection of $\mathbb{Y}$ onto the span of $\mathbb{X}$.
- ☒ **True** ☐ False If $\mathbb{X}$ is full column rank, there is exactly one $\hat{\theta}$ vector that minimizes the mean squared error.
- ☒ **True** ☐ False The prediction vector $\hat{\mathbb{Y}}$ is a linear combination of the columns of $\mathbb{X}$.

Solution: The first statement is false: the residual vector lives in $\mathbb{R}^n$, while $\vec{\theta}$ lives in $\mathbb{R}^p$ — these generally differ.

The second statement is true: OLS minimizes $\|\mathbb{Y} - \mathbb{X}\theta\|_2^2$, which is exactly orthogonal projection of $\mathbb{Y}$ onto the span of $\mathbb{X}$.

The third statement is true: full column rank makes $\mathbb{X}^\top \mathbb{X}$ invertible, giving the unique solution $\hat{\theta} = (\mathbb{X}^\top \mathbb{X})^{-1} \mathbb{X}^\top \mathbb{Y}$.

The fourth statement is true: by definition $\hat{\mathbb{Y}} = \mathbb{X}\hat{\theta} = \sum_j \hat{\theta}_j \cdot (\text{column } j)$, a linear combination of $\mathbb{X}$'s columns.

- (c) [2 Pts] Kevin first fits the following simple model to predict `response_time` using only `blocks_away`:

$$\text{response_time} = \hat{\theta}_0 + \hat{\theta}_1 \times (\text{blocks_away})$$

- (i) [1 Pt] In one sentence, interpret the meaning of $\hat{\theta}_0$ in this model.

Solution: $\hat{\theta}_0$ is the model's predicted `response_time` when `blocks_away`=0.

- (ii) [1 Pt] Kevin now one-hot encodes the categorical feature borough for which NYC borough Spiderman is currently in (which takes on values Manhattan, Brooklyn, or Queens), and adds two new features to his model, leaving out the Manhattan column:

$$\text{response_time} = \hat{\theta}_0 + \hat{\theta}_1 \times (\text{blocks_away}) + \hat{\theta}_2 \times (\text{Brooklyn}) + \hat{\theta}_3 \times (\text{Queens})$$

In one sentence, interpret the meaning of $\hat{\theta}_0$ in this **new** model.

Solution: $\hat{\theta}_0$ is now the predicted response_time when blocks_away= 0 and the call is in Manhattan specifically (the reference level, where both one-hot columns are 0)

- (d) [4 Pts] Kevin then tries a completely different loss function to predict response_time using two features:

- x_{i1} : blocks away from the emergency on call i
- x_{i2} : wind speed on call i

He defines the following regularized loss function:

$$L(\theta_1, \theta_2) = \frac{1}{2n} \sum_{i=1}^n (y_i - \theta_1 x_{i1} - \theta_2 x_{i2})^2 + \frac{\lambda}{2} (\theta_1^2 + \theta_2^2).$$

Kevin decides to minimize this loss using gradient descent.

- (i) [2 Pts] Derive $\frac{\partial L}{\partial \theta_1}$. Show your work.

Solution: For θ_1 :

$$\frac{\partial L}{\partial \theta_1} = -\frac{1}{n} \sum_{i=1}^n (y_i - \theta_1 x_{i1} - \theta_2 x_{i2}) x_{i1} + \lambda \theta_1.$$

He starts from $\theta_1^{(0)} = 0$, $\theta_2^{(0)} = 0$, uses a learning rate $\alpha = 0.1$, and a regularization term $\lambda = 0.5$. **Kevin uses new data to fit this model:**

$$\mathbb{X} = \begin{bmatrix} 1 & 3 \\ 2 & 1 \\ 4 & 2 \end{bmatrix}, \quad \mathbb{Y} = \begin{bmatrix} 10 \\ 15 \\ 20 \end{bmatrix}$$

(ii) [2 Pts] Use gradient descent to calculate $\theta_1^{(1)}$. Show your work.

Solution: At $t = 0$, $\theta_1^{(0)} = \theta_2^{(0)} = 0$, so $\hat{y}_i^{(0)} = 0$ and $y_i - \hat{y}_i^{(0)} = y_i$. From part (i) with $\theta_1^{(0)} = 0$:

$$\left. \frac{\partial L}{\partial \theta_1} \right|_{(0,0)} = -\frac{1}{n} \sum_{i=1}^n y_i x_{i1} = -\frac{1}{3} [(10)(1) + (15)(2) + (20)(4)] = -\frac{120}{3} = -40.$$

Applying the gradient descent update with $\alpha = 0.1$:

$$\theta_1^{(1)} = \theta_1^{(0)} - \alpha \left. \frac{\partial L}{\partial \theta_1} \right|_{(0,0)} = 0 - 0.1(-40) = 4.$$

(e) [2 Pts] Three different training paradigms are used to optimize a model. Each runs for **5 epochs**.

Paradigm	Number of Data Points
Paradigm 1 (Batch GD)	150
Paradigm 2 (Mini-batch GD)	200
Paradigm 3 (Stochastic GD)	10

Which ordering correctly ranks the paradigms by total number of gradient computations **across all epochs**?

Note: Paradigm 2 uses a batch size of 10.

- ☐ Paradigm 1 < Paradigm 2 < Paradigm 3
☒ **Paradigm 1 < Paradigm 3 < Paradigm 2**
☐ Paradigm 3 < Paradigm 1 < Paradigm 2
☐ Paradigm 2 < Paradigm 3 < Paradigm 1

Solution: Paradigm 1 (Batch GD): $1 \times 5 = 5$ gradients.

Paradigm 2 (Mini-batch GD): $\frac{200}{10} = 20$ mini-batches/epoch $\Rightarrow 20 \times 5 = 100$ gradients.

Paradigm 3 (SGD): $10 \times 5 = 50$ gradients.

Thus: Paradigm 1 < Paradigm 3 < Paradigm 2.

(f) [2 Pts] Select all statements below that are true regarding gradient descent.

- ☐ With an **infinite** number of updates and a **fixed** learning rate, gradient descent will always converge to a **global** minimum.
- ☐ With an **infinite** number of updates and a **fixed** learning rate, gradient descent will always converge to a **local** minimum.
- ☒ **The gradient descent algorithm can compute gradient updates using both convex and non-convex functions.**
- ☐ Consider the average of the estimated gradients computed over one epoch of stochastic gradient descent. This value is always equal to the true gradient of the loss surface at the initial values of the model parameters.

Solution: The first statement is false: GD is only guaranteed to reach the global minimum for convex surfaces; non-convex surfaces can trap it elsewhere.

The second statement is false: even a local minimum isn't guaranteed — with a fixed learning rate, GD can oscillate or diverge instead.

The third statement is true: GD is a general-purpose procedure and runs (mechanically) on both convex and non-convex losses, though it's only guaranteed optimal in the convex case.

The fourth statement is false: SGD updates θ after every point, so later points in the epoch are evaluated at already-changed parameters, not the fixed initial ones.

7 Hit Me With Your Best Shot [5 Pts]

Rohan wants to model his rate of perfect shots using the estimator:

$$\hat{p} = \frac{1}{n} \sum_{i=1}^n X_i$$

Let X_i be an indicator random variable that equals 1 if Rohan makes a perfect shot on trial i , and 0 otherwise, for $i = 1, \dots, n$. Each trial is independent, so $X_1, \dots, X_n$ are i.i.d. Bernoulli(p) random variables. Let the actual probability of Rohan making a perfect shot be an unknown constant p (i.e., $P(X_i = 1) = p$ for each i). For each subpart below, grading will be done based on the work you show in the boxes below.

- (a) [2.5 Pts] What is the variance of $\hat{p}$? **Express and simplify your answer in terms of n , $\hat{p}$, p , and any real number.**

Solution: Since $X_1, \dots, X_n$ are i.i.d. Bernoulli(p), each has $\text{Var}(X_i) = p(1 - p)$. Using independence to distribute the variance across the sum:

$$\text{Var}(\hat{p}) = \text{Var}\left(\frac{1}{n} \sum_{i=1}^n X_i\right) = \frac{1}{n^2} \sum_{i=1}^n \text{Var}(X_i) = \frac{1}{n^2} \cdot n \cdot p(1 - p) = \frac{p(1 - p)}{n}.$$

- (b) [2.5 Pts] What is the bias of this estimator: $\mathbb{E}[\hat{p}] - p$? **Express and simplify your answer in terms of n , $\hat{p}$, p , and any real number.**

Solution: Since $X_1, \dots, X_n$ are i.i.d. Bernoulli(p), each has $\mathbb{E}[X_i] = p$, so

$$\mathbb{E}[\hat{p}] = \mathbb{E}\left[\frac{1}{n} \sum_{i=1}^n X_i\right] = \frac{1}{n} \sum_{i=1}^n \mathbb{E}[X_i] = \frac{1}{n}(np) = p.$$

Therefore,

$$\text{Bias}(\hat{p}) = \mathbb{E}[\hat{p}] - p = p - p = 0.$$

Since the bias is 0, $\hat{p}$ is an unbiased estimator of p .

8 A Messi Affair [9 Pts]

The 2026 World Cup is underway, and Aadil wants to predict whether Messi's Argentina wins their next match. Let $y_i = 1$ if Messi wins match i , and $y_i = 0$ otherwise.

- (a) [3.5 Pts] Aadil wants to build a logistic regression model using design matrix $\mathbb{X}$ to predict whether Argentina wins. This matrix has two features: $\mathbb{X}_{[:,1]}$ (`kids_there`, a boolean indicator of whether one of Messi's kids is at the game) and $\mathbb{X}_{[:,2]}$ (`rest_days`, the number of rest days before the match).

He builds a logistic regression model using $\mathbb{X}_{[:,1]}$ and $\mathbb{X}_{[:,2]}$ to predict whether Argentina wins ($y_i = 1$) or loses ($y_i = 0$). The model uses the following parameter vector:

$$\vec{\theta} = [-2 \quad 2 \quad 1]^\top$$

and follows the standard logistic regression form.

For a particular match m , one of Messi's kids was there and the team had 0 rest days before the match.

- (i) [1.5 Pts] Compute the probability $\hat{p}$ that Argentina wins match m .

Solution:

$$\hat{p} = \frac{1}{1 + e^{-(\theta_0 + \theta_1 \mathbb{X}_{[m,1]} + \theta_2 \mathbb{X}_{[m,2]})}} = \frac{1}{1 + e^{-(-2 + 2(1) + 1(0))}} = \frac{1}{1 + e^{-0}} = 0.5$$

- (ii) [.5 Pts] Using a probability threshold of 0.45, does the model predict Argentina wins or loses match m ?

- ☐ Argentina loses
- ☒ **Argentina wins**
- ☐ On the boundary
- ☐ Not enough information

- (iii) [1.5 Pts] Derive the equation of the decision boundary using a probability threshold of 0.5. Express your answer in terms of $\mathbb{X}_{[:,1]}$ and $\mathbb{X}_{[:,2]}$. *Note: You may rearrange the equation so that constants are on either side. For example, $a^2 + b = 5$ would receive the same credit as $a^2 + b - 5 = 0$.*

Solution:

$$\hat{p} = \frac{1}{1 + e^{-z}} = 0.5 \Rightarrow z = 0, \text{ where } z = x^\top \vec{\theta}$$

$$-2 + 2\mathbb{X}_{[s,1]} + \mathbb{X}_{[s,2]} = 0 \Rightarrow 2\mathbb{X}_{[:,1]} + \mathbb{X}_{[:,2]} = 2$$

- (b) [1 Pt] Evaluate the following statements about logistic regression models.

- ☒ **True** ☐ False For any $t \geq 0$, the sigmoid function σ satisfies $\sigma(t) \in [0.5, 1)$.
- ☒ **True** ☐ False In binary classification, a logistic regression model can predict the probability of a datapoint belonging to class 0.
- ☐ True ☒ **False** For all linearly separable datasets, there exists a finite number of optimal solutions for model parameters.
- ☐ True ☒ **False** Let $\mathbb{X}$ be the design matrix. The quantity $\mathbb{X}^\top \mathbb{X}$ must be invertible in order to compute the optimal parameter vector.
- ☐ True ☒ **False** It is possible for the sigmoid function σ to output exactly $P(y_i = 1|x_i) = 1$ for some datapoint x_i with real-valued features.

Solution: The first statement is true. A sigmoid function's expression is $\sigma(z) = \frac{1}{1+\exp(-z)}$. Therefore, for all $z \geq 0$, $\sigma(z) \geq 0.5$.

The second statement is true. This is because the logistic regression model conventionally predicts the probability of a datapoint belonging to class 1, which is the complement of a datapoint belonging to class 0.

The third statement is false. On a linearly separable dataset, the training loss keeps decreasing as $\|\hat{\theta}\|$ grows without bound (pushing predicted probabilities toward exactly 0 or 1), so there is no finite optimal parameter vector at all — not a finite number of them.

The fourth statement is false. Unlike OLS, logistic regression has no closed-form solution requiring $\mathbb{X}^\top \mathbb{X}$ to be invertible — it's fit iteratively (e.g., via gradient descent).

The fifth statement is false. The sigmoid function's range is the open interval $(0, 1)$; it can get arbitrarily close to 1 but never equal it for any finite, real-valued input.

The sixth statement is true. $\sigma(x^\top \theta)$ is a nonlinear (sigmoidal) transformation of a linear combination of the features.

- (c) To evaluate Aadil's logistic regression model, he tests it on 6 new matches and compares predicted versus true outcomes. Results are shown below:

Match	y	$\hat{y}$	$\hat{P}_\theta(y = 1 x)$
1	1	1	0.75
2	1	1	0.65
3	0	1	0.55
4	1	0	0.48
5	1	0	0.35
6	0	0	0.20

As a reminder, $y = 1$ indicates Argentina won the match, and $\hat{y}$ is the model's predicted label at whatever threshold it currently uses.

- (i) [1 pt] What is the recall of the model on this data? Leave your answer as a fraction.

Solution: $FN = 2$ (Matches 4, 5). $\text{Recall} = \frac{TP}{TP+FN} = \frac{2}{4} = \frac{1}{2}$.

- (ii) [1 pt] Given the values in the table above, what are the minimum and maximum possible values of the threshold used to produce these predictions? *For full credit, your answer does not need to address whether the bounds are inclusive or exclusive.*

Solution: Matches predicted 1 have probabilities $\{0.75, 0.65, 0.55\}$ (minimum 0.55); matches predicted 0 have probabilities $\{0.48, 0.35, 0.20\}$ (maximum 0.48). The threshold must separate these two groups: (0.48, 0.55).

- (iii) [2.5 pts] What will happen to this logistic regression classifier's metrics if the classifier's threshold is lowered? Evaluate the following statements.

- ☐ True ☒ **False** Precision will never increase.
☒ **True** ☐ False Recall will never decrease.
☒ **True** ☐ False It is possible that the model will predict "Messi wins" more often.
☐ True ☒ **False** It is possible that the model's ROC-AUC will change.

Solution: Recall that $\text{Precision} = \frac{TP}{TP+FP}$, $\text{Recall} = \frac{TP}{TP+FN}$.

Lowering the classifier threshold can lead to increases in both True Positives and False Positives. As both the numerator and denominator are shifting, the precision can either increase or decrease as we tune the classifier threshold — so it is false that precision will *never* increase.

In the recall metric, the denominator never changes, as it is the number of data-points belonging to class 1. By lowering the classifier threshold, the number of true positives will either stay the same or increase, so recall will never decrease.

Lowering the threshold makes it easier for the model to predict a win, so it can only predict “Messi wins” the same or more often — never less, so it’s true that it’s possible for the model to predict wins more often.

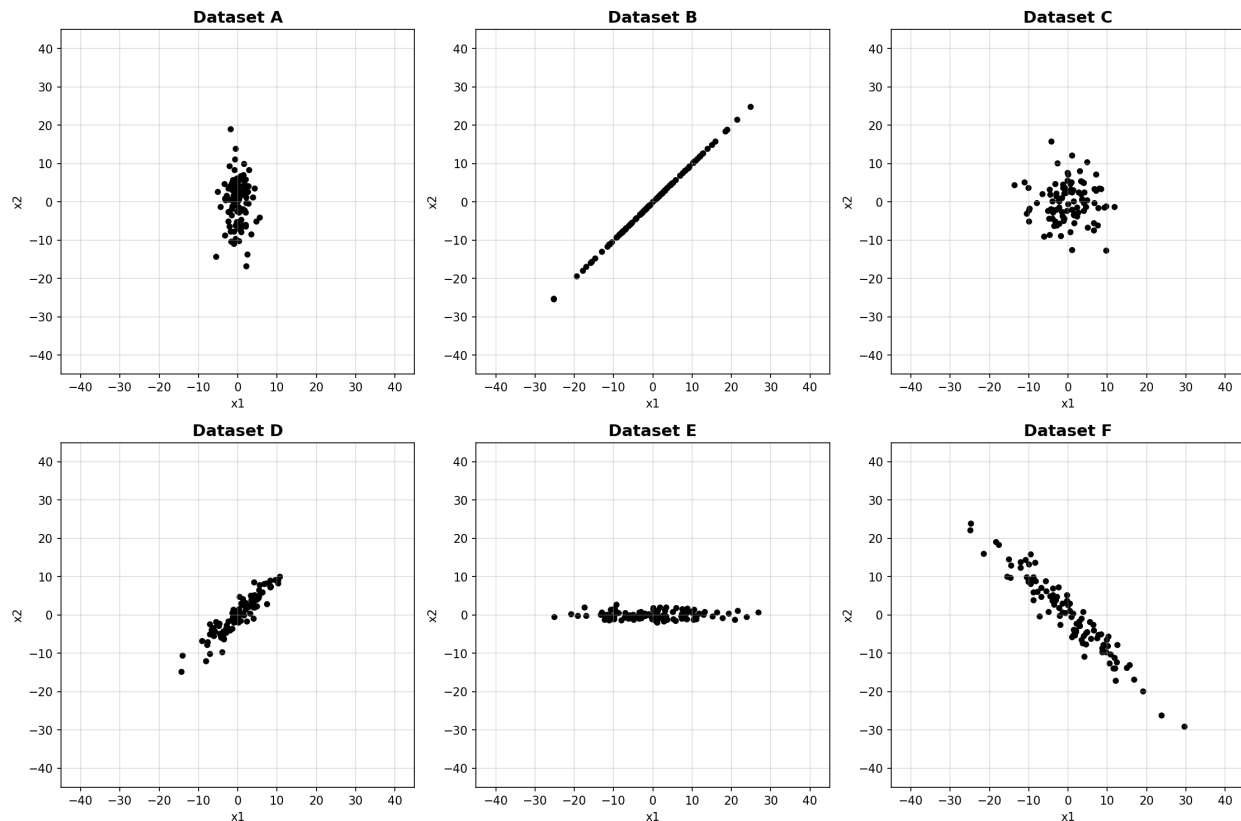
The model’s ROC-AUC is computed by sweeping over *all* thresholds at once, so changing a single threshold cannot change it — it’s false that ROC-AUC can change here.

Since precision can change (as shown above) and it’s one component of the F_1 score (the harmonic mean of precision and recall), the F_1 score can also change.

The confidence our model needs for predicting $\hat{y} = 1$ is decreased, not increased, by a lower threshold. A perfect predictor is defined as a model with an area under the ROC curve of 1; changing the threshold of a classifier doesn’t change its area under the ROC curve, so it cannot become a better or worse predictor this way.

9 [12 Pts] Dimensionality, Reducto!

In the parallel wizarding world, Professor Josh is looking at the performance of **100 students** in his TaDa 100 class, he records each student's **Charms Score** (x_1) and **Potions Score** (x_2) (**both mean-centered**). Below are six different possible versions of this dataset, labeled Dataset A through Dataset F. Each scatterplot is centered at the origin and drawn on the same axis scale.



(a) [2 Pts] Josh computes the covariance matrix of one of the six datasets above to be:

$$\Sigma = \begin{bmatrix} 4 & 0 \\ 0 & 36 \end{bmatrix}$$

Which scatterplot could this covariance matrix correspond to?

- ☒ **Dataset A**
- ☐ Dataset B
- ☐ Dataset C
- ☐ Dataset D
- ☐ Dataset E
- ☐ Dataset F

Solution: Dataset A. The covariance is 0, so the data is axis-aligned (no rotation). The variance along x_2 (36) is far larger than the variance along x_1 (4), so the cloud is stretched **vertically**.

(b) [2 Pts] Consider this ratio:

$$\frac{\text{variance of the 2nd latent feature}}{\text{variance of the 1st latent feature}}$$

For which of the following datasets is this ratio the **largest**?

- ☐ Dataset A
- ☐ Dataset B
- ☒ **Dataset C**
- ☐ Dataset D
- ☐ Dataset E
- ☐ Dataset F

Solution: Dataset C. Since PC1 always captures at least as much variance as PC2, this ratio is always ≤ 1 , with equality when the data is spread equally in every direction (an isotropic, circular cloud). Dataset C is the only roughly circular cloud, so it minimizes the ratio.

(c) [2 Pts] Josh takes one of the six datasets above, **samples 5** representative points, and computes the following (approximate) SVD of the resulting design matrix $\mathbb{X}$.

$$U \approx \begin{bmatrix} -0.71 & 0.10 \\ 0.24 & -0.55 \\ 0.47 & 0.60 \\ -0.35 & 0.30 \\ 0.35 & -0.40 \end{bmatrix} \quad S \approx \begin{bmatrix} 20 & 0 \\ 0 & 3 \end{bmatrix} \quad V^T \approx \begin{bmatrix} 0.71 & 0.71 \\ -0.71 & 0.71 \end{bmatrix}$$

(i) Which scatterplot could this SVD correspond to?

- ☐ Dataset A
- ☐ Dataset B
- ☐ Dataset C
- ☒ **Dataset D**
- ☐ Dataset E
- ☐ Dataset F

Solution: Dataset D. The first column of V^T , $(0.71, 0.71)$, points along the 45 direction so the problem simplifies to Dataset B or Dataset D. Dataset B is also tilted at 45, but S has two nonzero singular values here, so $\mathbb{X}$ has rank 2, while Dataset B's points fall exactly on a line (rank 1).

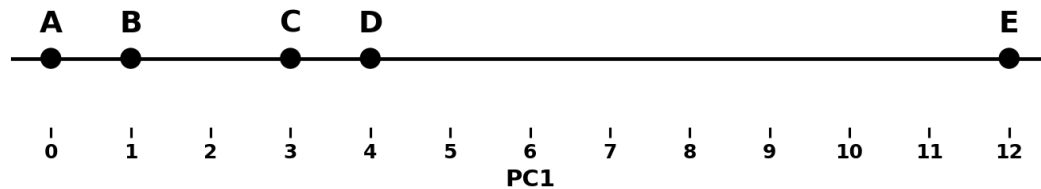
- (d) [2 Pts] Using the SVD given in part (c), what is the **component score** of the first principal component? In other words, what is the variance of the five sampled datapoints in the direction of the first principal component? You can leave your answer as an unsimplified algebraic expression (e.g., $(5 + 3)/2 + 1$). The matrix is repeated below for your convenience:

$$U \approx \begin{bmatrix} -0.71 & 0.10 \\ 0.24 & -0.55 \\ 0.47 & 0.60 \\ -0.35 & 0.30 \\ 0.35 & -0.40 \end{bmatrix} \quad S \approx \begin{bmatrix} 20 & 0 \\ 0 & 3 \end{bmatrix} \quad V^T \approx \begin{bmatrix} 0.71 & 0.71 \\ -0.71 & 0.71 \end{bmatrix}$$

Solution: The variance captured by a principal component is the corresponding eigenvalue of the covariance matrix, σ_j^2/n . There are $n = 5$ students, and the first singular value is $\sigma_1 = 20$, so the component score of the first principal component is

$$\frac{20^2}{5} = \frac{400}{5} = 80.$$

(e) [2 Pts] Cole projects a different sampling of points onto the first principal component.



Cole decides to use **K-medians clustering**. **K-medians clustering is exactly the same as K-means clustering, but it uses the median instead of the mean as the cluster centers.** He initializes Cluster 1's center at 0 and Cluster 2's center at 7.

Write the cluster center updates of K-medians clustering for this dataset. Write the updates in order and **until the algorithm converges**. We have provided a template for you to fill in. Each row represents one iteration of K-medians clustering.

Write the location of the updated Cluster 1 center and Cluster 2 center after each assignment step. **You do not have to use all rows.** If the algorithm has already converged (i.e., the centers don't change), please write NA in any boxes you do not use.

*Note: Remember Cole is using **K-medians clustering**, not K-means clustering. If you need to find the median of an even number of points, use the average of the two middle points.*

Step	Cluster 1 center	Cluster 2 center
Initialization	0	7
Step 1	1	8
Step 2	2	12
Step 3	NA	NA
Step 4	NA	NA

Solution: Step 1: with centers at 0 and 7, points A, B, C (positions 0, 1, 3) are closer to 0, while points D, E (positions 4, 12) are closer to 7. The new median of $\{A, B, C\}$ is 1; the new median of $\{D, E\}$ is $\frac{4+12}{2} = 8$.

Step 2: with centers at 1 and 8, points A, B, C, D (positions 0, 1, 3, 4) are closer to 1, while point E (position 12) is closer to 8. Since $\{A, B, C, D\}$ has an even number of points, its median is the average of the two middle values, $\frac{1+3}{2} = 2$; the median of $\{E\}$ is 12.

Step 3: with centers at 2 and 12, the assignment is unchanged ($\{A, B, C, D\}$ and $\{E\}$), so the centers don't move. The algorithm has converged, and Steps 3 and 4 are NA.

(f) [2 Pts] Which of the following statements are true?

- ☐ **True** ☐ False A negative silhouette score suggests that a point is, on average, closer to the points in the nearest neighboring cluster compared to the points in its own cluster.
- ☐ **True** ☐ False Agglomerative clustering will always converge to the same solution if re-run with the same parameters. *For this option, you can assume there are never exact ties in distances between clusters.*
- ☐ True ☐ **False** For a fixed dataset and a fixed value of K , K-means clustering always produces the same final cluster assignments, regardless of the initial placement of centroids.
- ☐ True ☐ **False** K-means optimizes the silhouette score.

Solution: A silhouette score is negative exactly when a point's average distance to points in the nearest neighboring cluster is smaller than its average distance to points in its own cluster — a sign it may be assigned to the wrong cluster. Unlike K-means, agglomerative clustering has no random initialization: given a fixed linkage criterion and no ties, the greedy merge order is entirely determined by the pairwise distances, so re-running it changes nothing.

K-means, on the other hand, **is** randomly initialized, so different starting centroids can converge to different final assignments and different local optima — it is only guaranteed to converge to *a* local optimum, not the global one. K-means minimizes within-cluster sum of squared distances (inertia), not the silhouette score, so those two objectives can disagree. Finally, K itself is a hyperparameter that must be chosen before running the algorithm; unlike agglomerative clustering, K-means cannot simply be "cut" at a different number of clusters after the fact.

Please state any relevant assumptions in the box below (Optional).

You are done with the final- Congratulations!

Draw your favorite DATA 100 memory so far!